Initial assignment across the partner episode — all seeds

Seed 1 · Seed 2 · Seed 3 · Seed 4 · Seed 5 · Mean of 5 seeds

v1_matched | v2 | v3

MLP Ego-red assignment (%)

RNN Ego-red assignment (%)

RNN single-partner Ego-red assignment (%)

RNN no influence Ego-red assignment (%)

Round within partner episode

Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Red means ego takes red, partner takes blue. Blue assignment percentage = 100 − red percentage.

Final assignment across the partner episode — all seeds

Seed 1 | Seed 2 | Seed 3 | Seed 4 | Seed 5 | Mean of 5 seeds

v1_matched | v2 | v3

MLP Ego-red assignment (%)

RNN Ego-red assignment (%)

RNN single-partner Ego-red assignment (%)

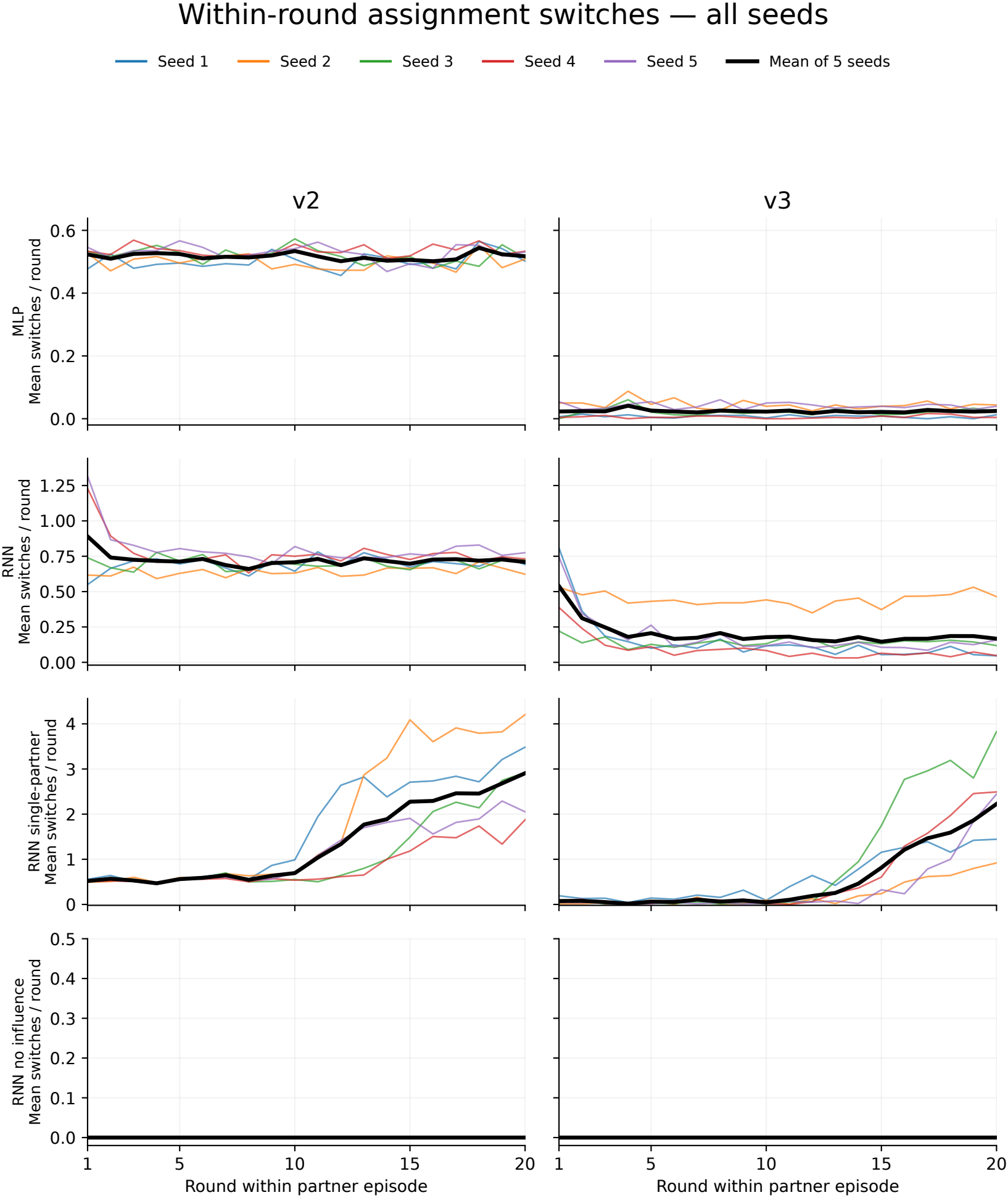
RNN no influence Ego-red assignment (%)

0 | 20 | 40 | 60 | 80 | 100

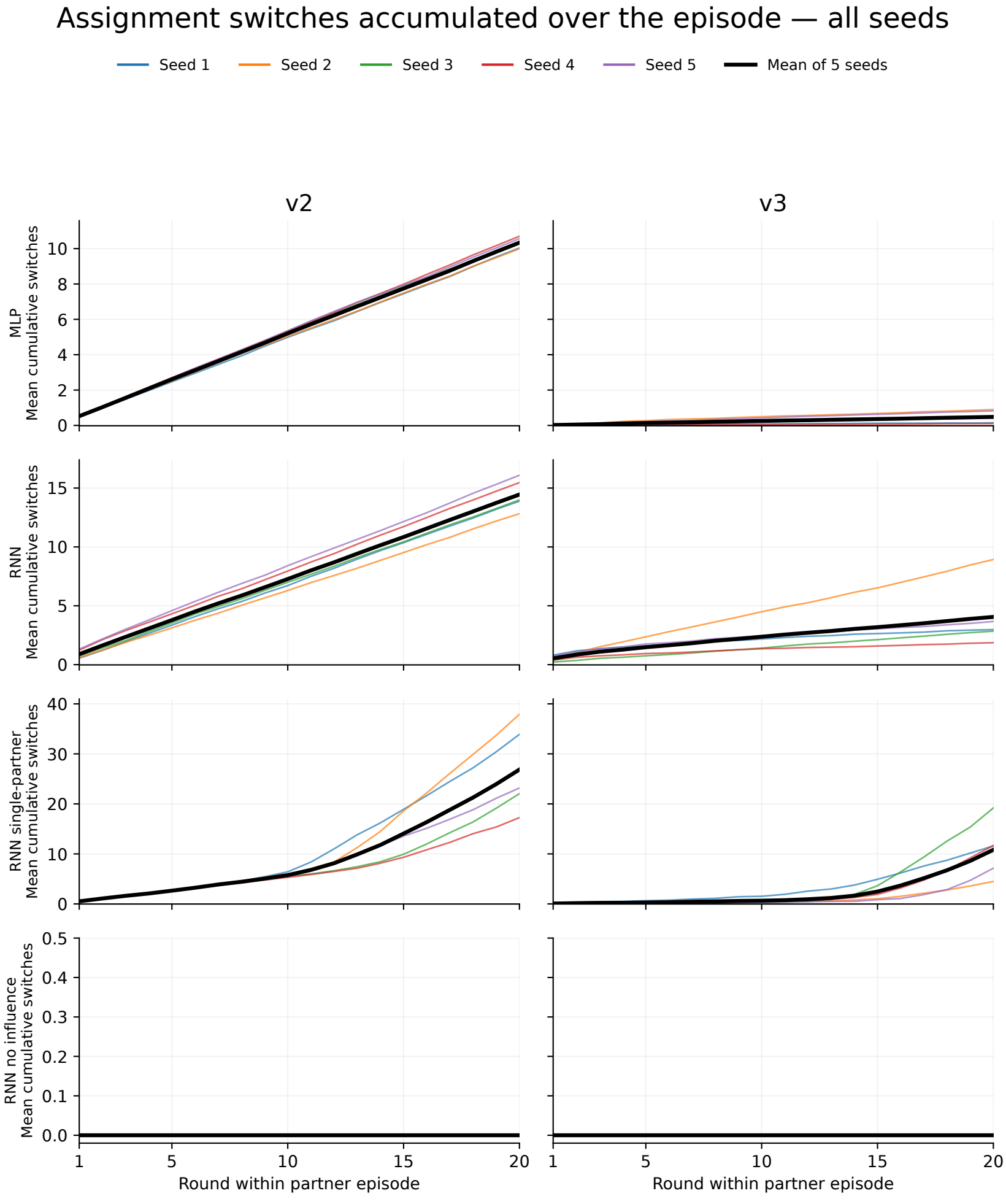
1 | 5 | 10 | 15 | 20

Round within partner episode

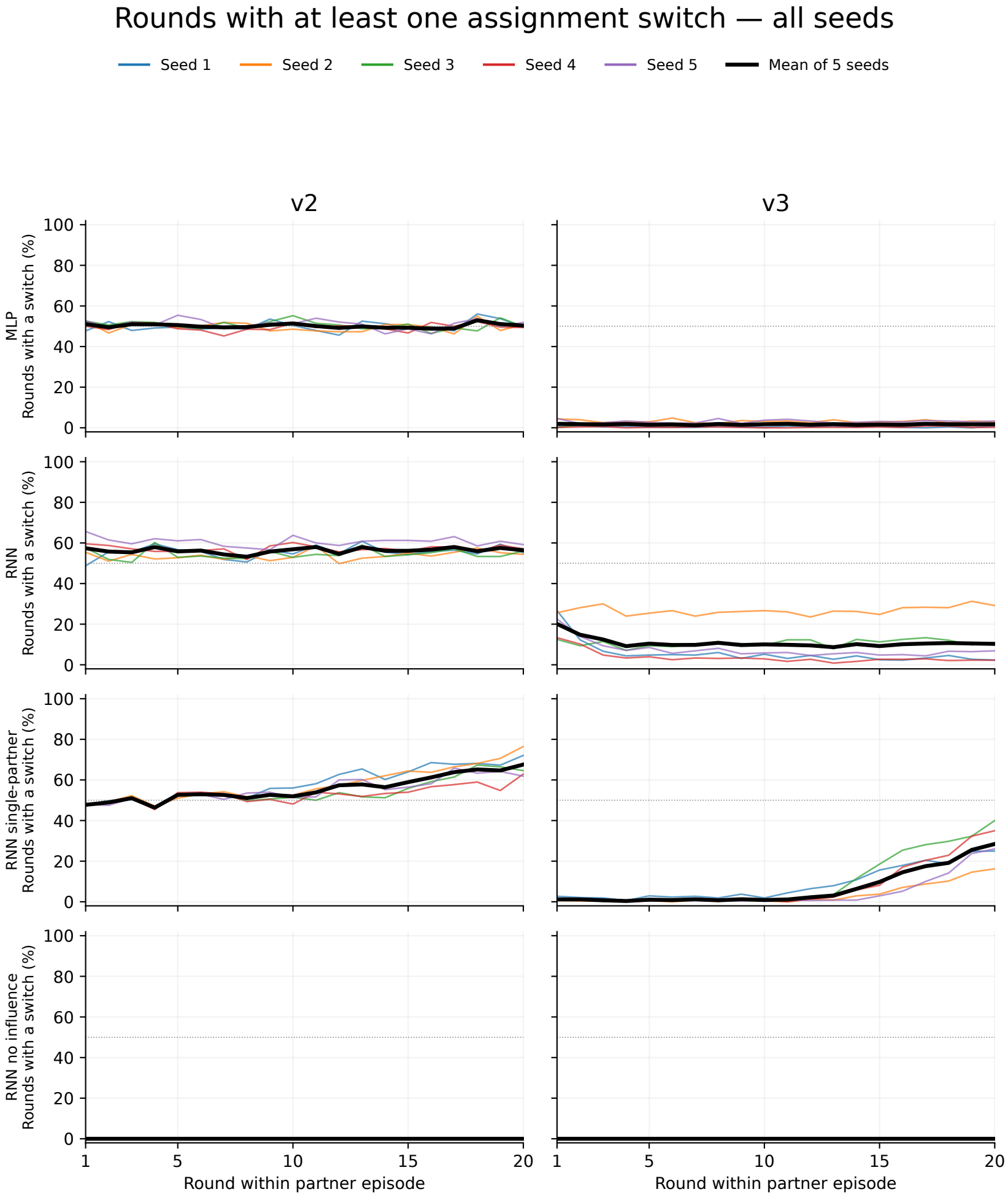
Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Red means ego takes red, partner takes blue. Blue assignment percentage = 100 − red percentage.



Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.



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Switches are actual within-round assignment changes; round resets and initial choices are excluded.

Final assignment matches partner's faster task — all seeds

Seed 1 · Seed 2 · Seed 3 · Seed 4 · Seed 5 · Mean of 5 seeds

v1_matched | v2 | v3

MLP Partner on faster task (%)

RNN Partner on faster task (%)

RNN single-partner Partner on faster task (%)

RNN no influence Partner on faster task (%)

0 20 40 60 80 100

1 5 10 15 20

Round within partner episode

Every line averages complete rounds over all partner profiles for one training seed; black is the equal-seed mean.
Switches are actual within-round assignment changes; round resets and initial choices are excluded.

Final assignments by partner ability

Partner faster at red (speed-matching ego-red target: 0%)
Partner faster at blue (speed-matching ego-red target: 100%)

v1_matched | v2 | v3

MLP Final ego-red assignment (%)
RNN Final ego-red assignment (%)
RNN single-partner Final ego-red assignment (%)
RNN no influence Final ego-red assignment (%)

0 20 40 60 80 100

1 5 10 15 20
Round within partner episode

Thin lines: all five individual training seeds; thick lines: equal-seed means.
Partner color is complementary to ego assignment. Speed matching is a behavioral diagnostic, not a geometry-aware optimal policy.

Assignment behavior: training seed 1

MLP · RNN · RNN single-partner · RNN no influence

v1_matched | v2 | v3

Initial ego-red assignment (%)

Final ego-red assignment (%)

Mean switches per round

Assignments fixed within each round

Final partner on faster task (%)

Round within partner episode

Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

Assignment behavior: training seed 2

MLP · RNN · RNN single-partner · RNN no influence

v1_matched | v2 | v3

Initial ego-red assignment (%)

Final ego-red assignment (%)

Mean switches per round

Assignments fixed within each round

Final partner on faster task (%)

Round within partner episode

Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

Assignment behavior: training seed 3

MLP · RNN · RNN single-partner · RNN no influence

v1_matched | v2 | v3

Initial ego-red assignment (%)

Final ego-red assignment (%)

Mean switches per round

Assignments fixed within each round

Final partner on faster task (%)

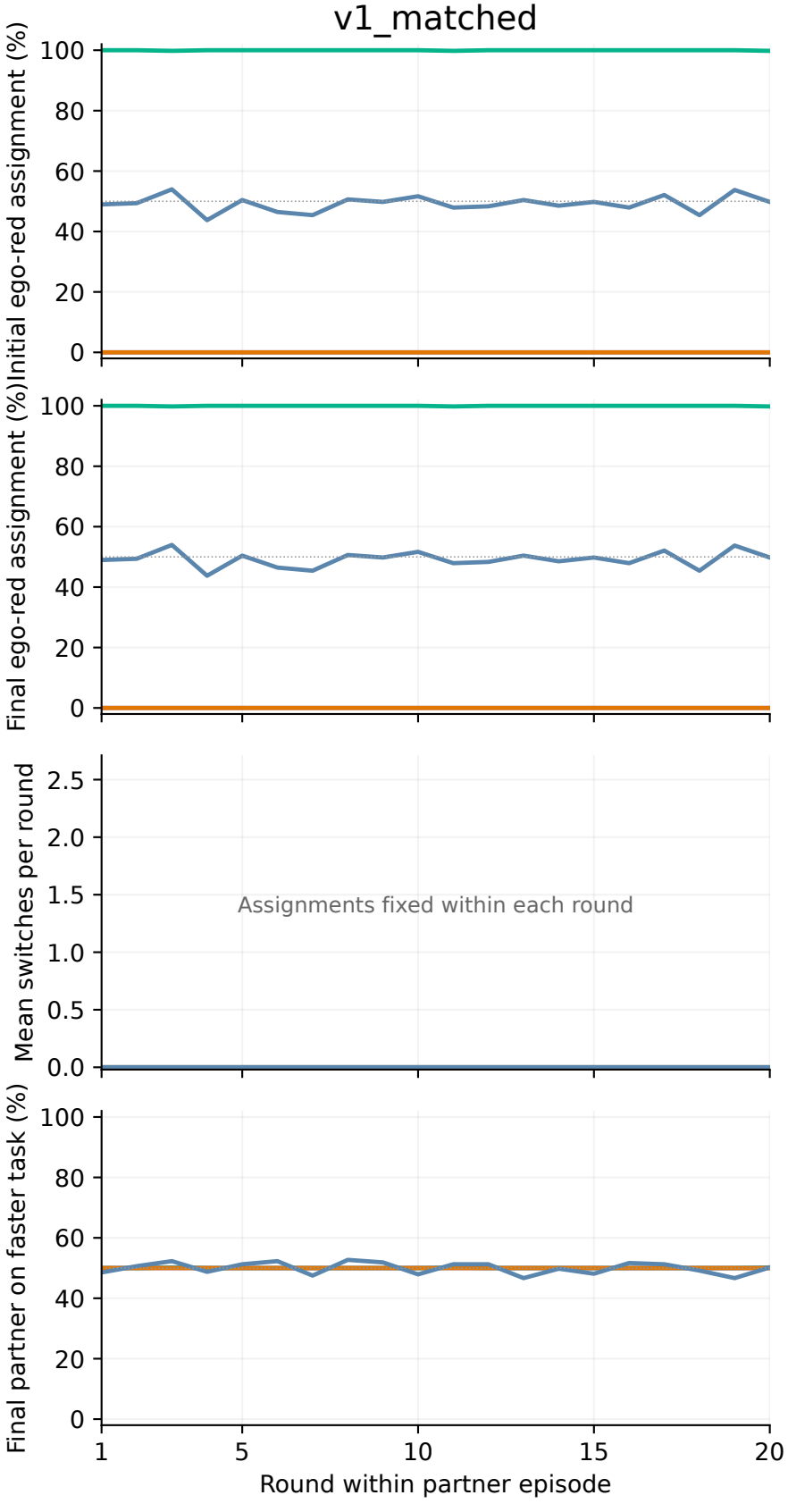
Round within partner episode

Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

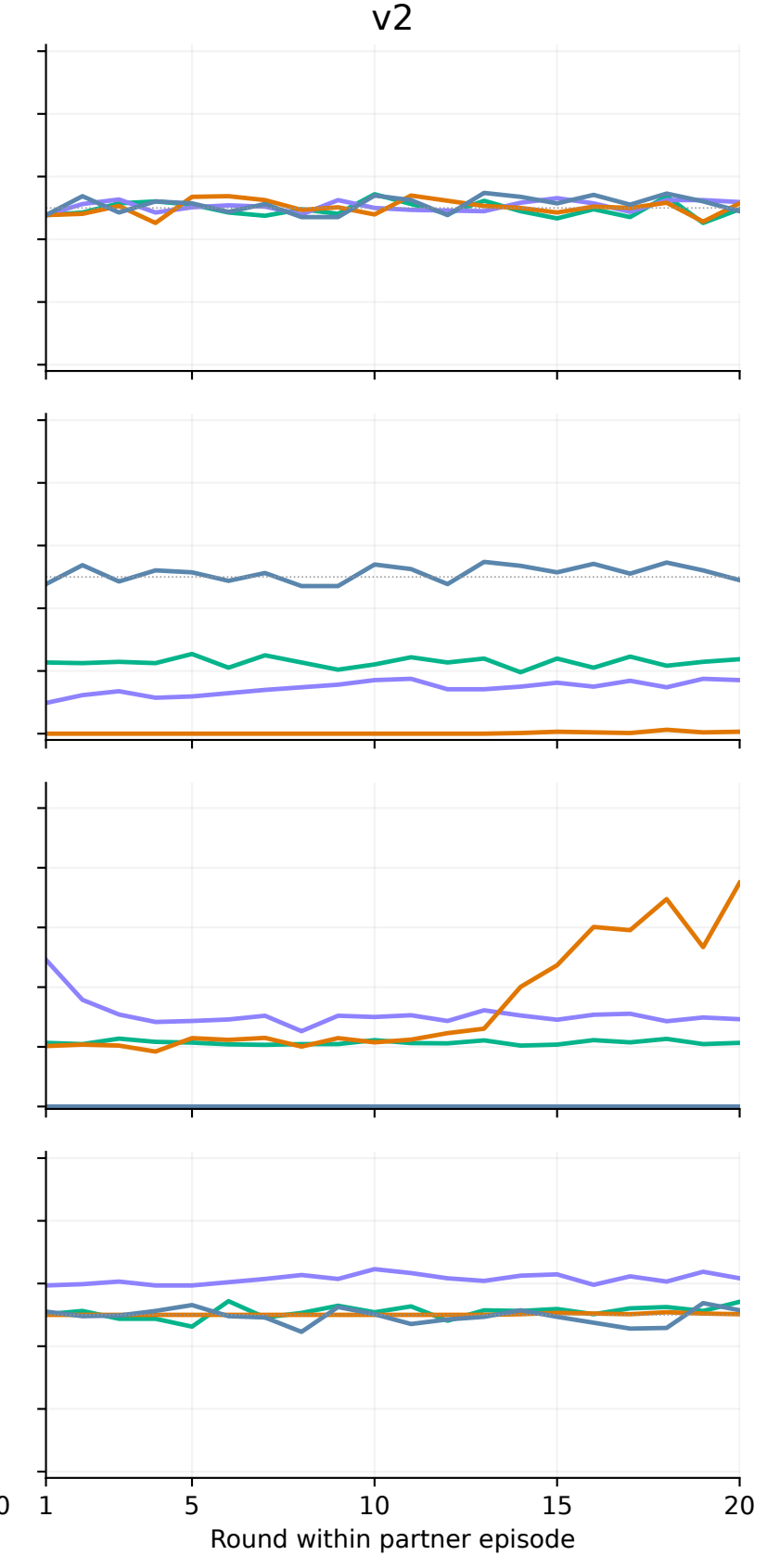
Assignment behavior: training seed 4

MLP · RNN · RNN single-partner · RNN no influence

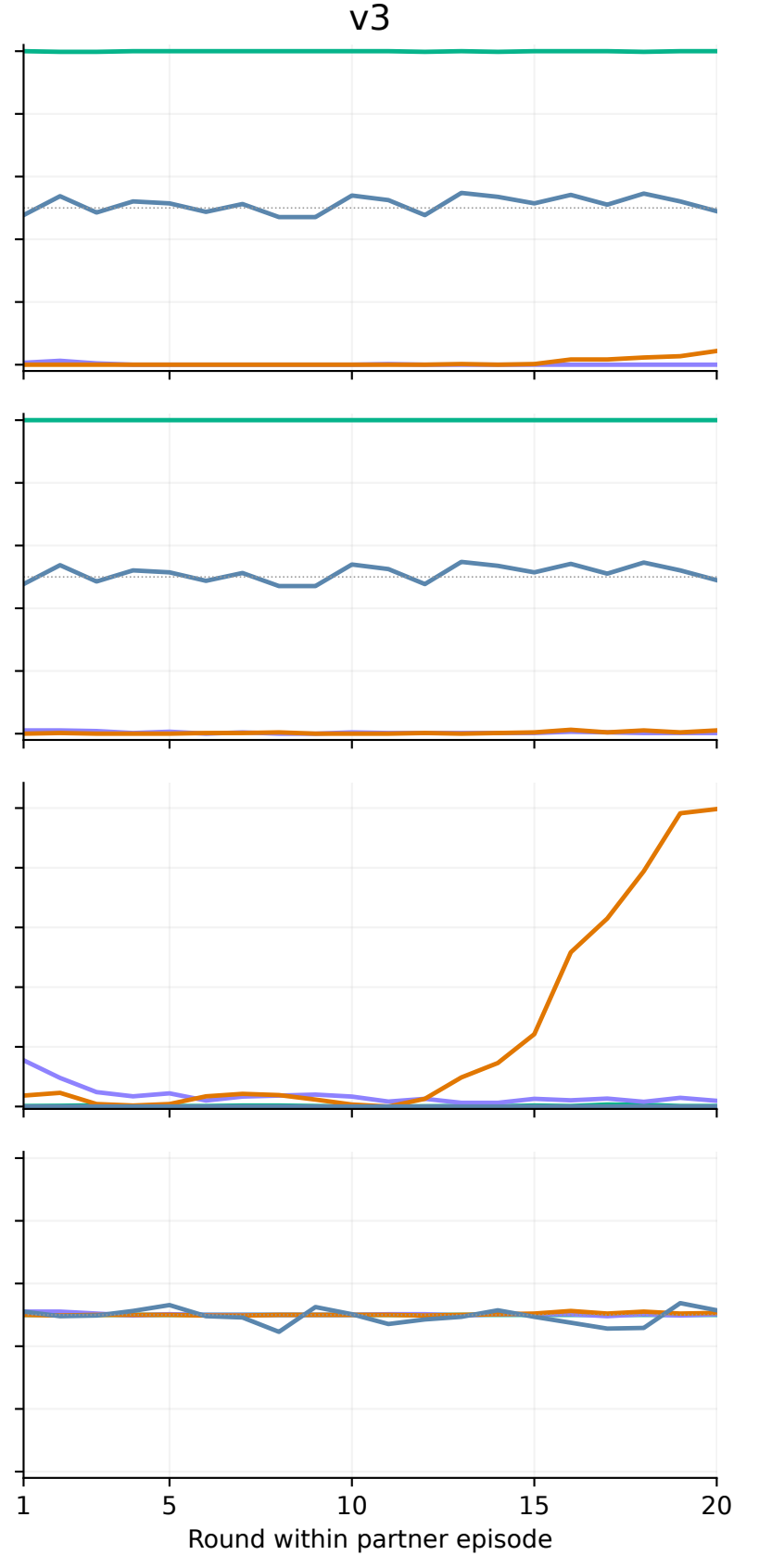
v1_matched



v2



v3



Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.

Assignment behavior: training seed 5

MLP · RNN · RNN single-partner · RNN no influence

v1_matched

v2

v3

Initial ego-red assignment (%)

Final ego-red assignment (%)

Mean switches per round

Assignments fixed within each round

Final partner on faster task (%)

Round within partner episode

Red assignment = ego red / partner blue. V2 initial assignments are random; V3 initial assignments are learned when influence is enabled.
No-influence curves show realized assignments, not voluntary choices. Faster-task matching is descriptive, not a full optimality score.